

Universal, Hard-Routed and Soft Mixture-of-Experts Image Restoration, and Style-Conditioned Face-to-Sketch Generation

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Abstract—This report presents four related generative systems built on a single data pipeline. Clean images of the Oxford-IIIT Pet dataset are corrupted at load time by one of three degradations—salt-and-pepper noise, Gaussian blur or rectangular occlusion—or left clean, and three systems of increasing routing sophistication learn to invert them. The first is a universal convolutional denoising autoencoder with a genuine compressed bottleneck, which is never told which corruption it is facing. The second replaces one shared model by a four-class corruption classifier that routes each image to one of three specialist autoencoders, with an identity bypass for clean inputs. The third makes that routing differentiable: a temperature-scaled softmax gate, initialised from the classifier, assigns continuous weights to the identity branch and the three specialists, and the whole system is fine-tuned jointly under a loss that combines reconstruction, structural similarity, corruption classification and a routing-balance regulariser. The fourth system is a conditional GAN that translates a facial photograph into a pencil sketch in one of three artist styles of the FS2K dataset, using a U-Net generator whose decoder is modulated by a learned style embedding and a PatchGAN discriminator conditioned by projection. Every model is tuned with Optuna, tracked with Weights & Biases, exported to ONNX with a verified parity check, and served through a FastAPI backend behind a React interface that runs under Docker Compose. **[TODO: one or two sentences of headline quantitative results once all four systems have been trained: Task 1 PSNR/SSIM per corruption, Task 2 classifier macro-F1 and routing gain, Task 3 soft-routing gain over hard routing, Task 4 per-style SSIM/LPIPS]**

Index Terms—image restoration, denoising autoencoder, mixture of experts, conditional generative adversarial networks, hyperparameter optimisation, ONNX

I. INTRODUCTION

Image restoration systems are usually built one degradation at a time. A denoiser is trained on noise, a deblurring network on blur, an inpainting network on holes, and the choice between them is left to whoever runs the pipeline. That arrangement is convenient for research but awkward in deployment, where the degradation afflicting an incoming image is rarely announced. Two families of answers have emerged. One trains a single model across degradations and relies on its capacity to cover all of them [1], [2], [3]; the other keeps specialised components and adds a mechanism that decides which to use [4], [5].

This assignment asks for both, and for the step that connects them. Using one fixed data pipeline we build three restoration

systems that differ only in how the knowledge of the corruption enters the computation.

- **Task 1** is a universal denoising autoencoder. It receives a corrupted image and nothing else: no label, no severity, no mask. All four input conditions—clean, salt-and-pepper, blur, occlusion—must be handled by one encoder, one compressed latent representation and one decoder.
- **Task 2** makes the corruption explicit. A convolutional classifier predicts which of the four conditions holds, and that discrete decision selects one of three specialist autoencoders, or an identity bypass when the image is judged clean. The system is evaluated twice: with the true label from the test manifest (*oracle routing*, which isolates how good the specialists are) and with the classifier’s own prediction (*predicted routing*, which is the operational system).
- **Task 3** replaces the `argmax` with a temperature-scaled softmax. The gate is the Task 2 classifier, the experts are the Task 2 specialists, and the output is their weighted sum. Because that sum is differentiable, the gate and the experts can be trained jointly on the final reconstruction error, with a balance regulariser to keep the gate from collapsing onto a single branch.

Task 4 is a different problem on a different dataset: paired image-to-image translation with a categorical condition. Given a facial photograph and one of three sketch styles, a conditional GAN must produce the sketch that the corresponding artist would have drawn.

Three themes run through the report. The first is *evidence*: every architectural choice—where the bottleneck sits, whether skip connections are allowed, which loss, which upsampling operator, which balance regulariser, how the style condition is injected—is argued from published results and, where the literature is ambiguous, measured. The second is *honesty about what the numbers mean*: PSNR is capped for identical images, clean inputs through an identity bypass are exact rather than merely excellent, FID is biased at small sample sizes, and the FS2K test set contains only 46 images of one style. Each of these is stated where the corresponding result is presented rather than buried. The third is *deployability*: all five inference models are exported to ONNX and checked against their

PyTorch originals, and the complete application starts with a single Docker Compose command.

The implementation, including the data preparation, training, search, evaluation and export scripts, the backend and the frontend, is at <https://github.com/fsdev87/generative-ai-A1>. A demonstration video is at [TODO: YouTube link to the 5–7 minute demonstration video].

II. RELATED WORK

A. Denoising autoencoders and restoration networks

The denoising autoencoder was introduced as a representation learner: an autoencoder trained to reconstruct a clean input from a stochastically corrupted one cannot settle for the identity and must capture the structure of the data [6], [7]. The corruption is applied during training only, which is exactly the runtime scheme used here. The compressed central layer is what makes the representation compact [8], and convolutional encoder-decoder stacks make the idea practical on images [9], [10]. DnCNN showed that one network can cover a *range* of noise levels rather than one model per level [1], the precedent for a single model covering several corruption families.

Two architectural traditions meet in Task 1. U-Net concatenates every encoder stage into the mirrored decoder stage [11], and RED-Net adds symmetric skip connections to a very deep restoration network, reporting both faster convergence through better gradient flow and better detail recovery because the skips “pass image details from convolutional layers to de-convolutional layers” [12]. That second mechanism is precisely what a bottleneck is supposed to forbid: a full-resolution skip lets the decoder reconstruct detail the latent never encoded. Context Encoders take the opposite route for inpainting, using a narrow bottleneck and no skips, because occluded content must be synthesised from context rather than copied [13]. Section IV measures this trade-off rather than assuming it.

B. Routing and all-in-one restoration

RL-Restore assembles a toolbox of small, differently specialised CNNs and learns by reinforcement learning which tools to apply and in what order [4]; Path-Restore extends the idea to spatially varying routing, selecting a path per image region with a difficulty-regulated reward [5]. Both report that a set of small specialists plus a routing policy is more parameter-efficient than one large network—the hypothesis Task 2 tests in its simplest form, with a single image-level decision and a supervised classifier instead of a learned policy. The opposite design removes the router entirely: AirNet infers the degradation with a contrastive encoder that then guides restoration [2], and PromptIR encodes degradation-specific information in lightweight learned prompts that modulate a single network [3]. These all-in-one models are the state of the art for the setting of Task 1; they are trained on far more data with far larger backbones, and no comparison with them is claimed here.

C. Mixtures of experts

The mixture of experts originates with Jacobs et al. [14], who distinguish a *cooperative* error function, in which the target is compared with a blend $\sum_i p_i o_i$ of expert outputs, from a *competitive* one, $\sum_i p_i \|d - o_i\|^2$, in which each expert must produce the whole output. They argue that the cooperative form “does not encourage localization”, because each expert only has to cancel the residual left by the others, and that it “tends to lead to solutions in which many experts are used for each case”. Jordan and Jacobs later gave the construction its statistical form as a mixture model trained by EM [15]; Yuksel et al. survey two decades of variants [16].

The modern, large-scale form adds sparsity. Shazeer et al. introduce noisy top- k gating and observe the central pathology: “the gating network tends to converge to a state where it always produces large weights for the same few experts”, an imbalance that is self-reinforcing because favoured experts are trained faster and therefore chosen more often [17]. Their remedy is a pair of soft constraints on batch statistics, discussed quantitatively in Section VI. Switch Transformers simplify the two losses into one [18], [19], and Soft MoE removes the problem structurally: every expert slot receives a weighted average of all tokens, so the model is “immune to token dropping and expert unbalance” and needs no auxiliary balance loss at all [20]. The system in Task 3 is dense and soft in the same sense—all four branches always run—but small enough that the gate, not the capacity, is the object of study. The wider framing is conditional computation, in which each input traverses only part of the network [21], [22]; Eigen et al. already observed the gating-collapse failure mode and addressed it with a hard constraint early in training, and Bengio et al. with a soft constraint on the batchwise average of each gate—the family to which the regulariser used here belongs.

D. Conditional image-to-image translation

Conditional GANs supply the condition to both generator and discriminator [23], [24]. pix2pix is the canonical paired translation model: a U-Net generator, a 70×70 PatchGAN discriminator, and an L_1 term with weight $\lambda = 100$ added to the adversarial loss, L_1 rather than L_2 because it “encourages less blurring” [25]. Isola et al. also report that removing the conditioning from the discriminator makes the generator collapse to nearly the same output regardless of input, which is direct evidence for conditioning both networks.

How a *categorical* condition should enter is a design question with several published answers. The simplest spatially replicates a label map and concatenates it to the input [26]. Conditional instance normalisation learns a per-condition scale and shift, letting a single network render many styles [27], a mechanism generalised as FiLM [28], [29] and applied to language-driven vision [30]; AdaIN extends it to arbitrary, unseen styles [31]. On the discriminator side, the auxiliary classifier of AC-GAN [32] competes with the projection discriminator, which adds an inner product between the embedded condition and the discriminator’s feature vector and

is motivated by the role of the condition in the underlying probabilistic model [33]. Section VII explains, and measures, why this work conditions the generator by FiLM and the discriminator by projection.

Face photo-to-sketch synthesis itself has a long history, from multiscale Markov random fields on the CUHK datasets [34] to the FS2K benchmark used here [35].

III. DATASET PREPARATION

Tasks 1–3 use the Oxford-IIIT Pet dataset [36], [37]: 7,349 photographs of 37 cat and dog breeds. Task 4 uses FS2K [35], [38], 2,104 photograph-sketch pairs in three artist styles. All data preparation is implemented in `src/data/` and the resulting splits and manifests are committed to the repository, so every run of every task sees identical data.

A. Splits

The official `trainval` list of the Oxford-IIIT Pet dataset contains 3,680 images and the official `test` list 3,669. As the brief requires, the development portion is divided 80/20 with random seed 42, giving **2,944 training** and **736 validation** images; the official test set of **3,669** images is left untouched until the final evaluation. All three splits are disjoint and all 37 breeds appear in each: the training split holds 71–91 images per breed (953 cats, 1,991 dogs), the validation split 9–29 per breed (235 cats, 501 dogs) and the test split 88–100 per breed (1,183 cats, 2,486 dogs). Breed labels are not used by any model—the clean image is the reconstruction target—but the per-breed counts confirm that the random split did not concentrate any breed in one partition.

Every image is converted to RGB and resized to 128×128 with bicubic resampling and no crop, then cached once as a `uint8` array so that training never re-decodes a JPEG. Pixel values are scaled to $[0,1]$ and stored as `float32` in NCHW order; all restoration models end in a sigmoid.

B. Corruption configuration

Four input conditions are used, in the fixed order (`clean`, `salt`, `blur`, `occlusion`) that indexes the classifier logits, the mixture weights and every table in this report; index 0 is always the clean or identity branch. Table I gives the training ranges, from which severities are sampled uniformly at training time, and the three fixed severity levels used for testing.

Three details of the implementation (`src/data/corruptions.py`) matter for the results.

Salt-and-pepper noise selects each pixel independently with probability p and sets all three channels to 0 or 1 with equal probability, so the corruption is a pure impulse: a hit pixel carries no information about its clean value.

Gaussian blur is separable with reflect padding and matches `torchvision's gaussian_blur` to within 10^{-5} , verified by a unit test. The kernel is truncated to k taps, and this truncation has a consequence that was discovered during data

preparation and changed how severity is labelled. The standard deviation of the kernel that is actually applied,

$$s(k, \sigma) = \left(\sum_{i=0}^{k-1} g_i \left(i - \frac{k-1}{2} \right)^2 \right)^{1/2}, \quad (1)$$

in which g denotes the normalised k -tap Gaussian kernel, saturates as σ grows, because a wide Gaussian truncated to k taps flattens towards a box filter whose standard deviation is bounded by $\sqrt{(k^2 - 1)/12}$. At $k = 3$ that bound is 0.816 px, so the configuration (3, 2.5) has strength 0.805 px—*milder* than the medium test level (5, 1.5) at 1.195 px, although its σ is larger. Ranking blur severity by σ alone is therefore wrong. Severity labels are instead assigned by binning $s(k, \sigma)$ with edges at the midpoints between the three test levels, 0.921 and 1.480 px. Regenerating the validation manifest under this rule left every corruption setting bit-identical and corrected **86 of its 184** blur severity labels.

Occlusion places n non-overlapping black rectangles whose union covers the target fraction to within one percentage point. The target area is divided among the rectangles by a Dirichlet draw, each rectangle receives a random aspect ratio in $[1/2, 2]$, and positions are rejected until no two overlap; the achieved coverage is recomputed from the union mask and resampled if it leaves the required range, so the stored coverage is the true one rather than the request.

A consequence of these definitions is worth stating before any result is reported: **all three test “high” levels sit at the upper edge of the training ranges**. Salt-and-pepper high uses $p = 0.15$, exactly the top of the training range; occlusion high covers 35%, exactly the top; and blur high, (7, 2.5) at strength 1.765 px, is the single strongest blur the training distribution can produce, reachable only at $k = 7$ with σ near its maximum. Under uniform sampling of (k, σ) only about 15% of training blurs fall in the “high” band at all, and none exceeds the test level. The high severities are thus the sparsest part of the training distribution, and the weakest results are expected there—not because the models fail, but because the brief’s test protocol probes the edge of what training covers.

C. Runtime corruption and deterministic manifests

Training corruptions are generated *at load time*: each access to a training image draws a fresh condition and a fresh severity, applies the corruption in NumPy and returns the pair (corrupted input, clean target) together with the integer label that produced it. Nothing corrupted is ever written to disk, so an image is seen under a different degradation in every epoch, which acts as a strong data augmentation and provides the classifier of Task 2 with free, perfectly accurate labels. A random horizontal flip is applied to the clean image before corruption.

Validation and test corruptions must instead be identical across runs, models and machines. They are therefore generated once into JSON *manifests* that store, for each entry, the corruption type, the severity level, every parameter (p ; k and σ ; the number of rectangles, achieved coverage and exact rectangle coordinates) and the random seed. Applying a stored

TABLE I

CORRUPTION CONFIGURATION. TRAINING SEVERITIES ARE SAMPLED FROM THE RANGES ON EVERY IMAGE LOAD; THE TEST SEVERITIES ARE FIXED BY THE BRIEF AND STORED IN THE TEST MANIFEST. BLUR STRENGTH IS THE STANDARD DEVIATION IN PIXELS OF THE KERNEL THAT IS ACTUALLY APPLIED AFTER TRUNCATION TO k TAPS.

Condition	Training range (sampled per load)	Test levels (low / medium / high)	Test strength
Clean	image passed through unchanged	—	—
Salt-and-pepper	$p \sim \mathcal{U}(0.02, 0.15)$; hit pixels set to 0 or 1 equiprobably	$p = 0.03 / 0.08 / 0.15$	—
Gaussian blur	$k \in \{3, 5, 7\}$ uniformly, $\sigma \sim \mathcal{U}(0.5, 2.5)$	$(3, 0.7) / (5, 1.5) / (7, 2.5)$	0.647 / 1.195 / 1.765 px
Rectangular occlusion	$n \in \{1, 2, 3\}$ rectangles covering $\mathcal{U}(0.10, 0.35)$ of the area	$n = 1, 2, 3$ at 10% / 20% / 35%	—

TABLE II

VALIDATION MANIFEST (736 ENTRIES, SEED 1001): ONE DETERMINISTIC CORRUPTION PER VALIDATION IMAGE, BALANCED ACROSS THE FOUR CONDITIONS.

Condition	Low	Medium	High	Total
Clean	(no severity)			184
Salt-and-pepper	46	97	41	184
Gaussian blur	92	62	30	184
Rectangular occlusion	36	91	57	184
Total				736

specification is fully deterministic, so the manifests reproduce the corrupted images exactly without storing them.

The **validation manifest** (seed 1001) assigns *one* condition to each of the 736 validation images, balanced across the four classes: exactly 184 entries each of clean, salt-and-pepper, blur and occlusion. Severities are drawn from the training ranges, so the validation distribution matches training. Table II gives the resulting severity composition. The distribution across levels is not uniform, and for a well understood reason: severity is binned on a physical quantity (p , covered fraction, blur strength) whose uniform sampling does not produce uniform bins. For blur the effect is strongest, since $k = 3$ can never reach the medium band at all.

The **test manifest** (seed 2002) is exhaustive instead of sampled: every one of the 3,669 test images appears in ten variants—clean, plus each of the three corruptions at each of the three fixed severity levels—giving **36,690 entries**, exactly 3,669 per (type, level) group. Because every image appears under every condition, comparisons across corruptions and severities are paired, and because the test set is balanced by construction, accuracy and micro-averaged F1 coincide for the Task 2 classifier. The achieved occlusion coverage across the test manifest is tightly controlled: mean 0.1000, 0.2000 and 0.3500 for the three levels, with extremes of 0.0987–0.1016, 0.1972–0.2028 and 0.3462–0.3544.

Figure 1 shows the four conditions at the three fixed test severities.

D. Class-balanced sampling

Wherever batches must be balanced—the Task 2 classifier and the Task 3 mixture—a custom batch sampler emits batches of (index, condition) pairs containing exactly $B/4$ images of each condition, reshuffling the images every epoch and dropping the final incomplete batch; batch sizes are therefore

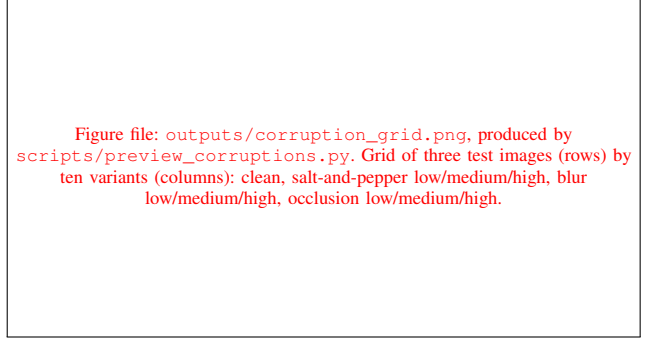


Fig. 1. The four input conditions at the three fixed test severities. Every test image appears in all ten variants in the test manifest.

multiples of four. Because the condition is drawn by the sampler rather than by the dataset, perfect balance costs nothing: there is no minority class to oversample. This is the strongest form of the remedy that Buda et al. find dominant for class imbalance in CNNs, namely oversampling to the point where the imbalance is eliminated entirely [39]. It also has a second consequence that Section VI depends on: in an exactly balanced batch, a perfectly routing gate assigns mean weight exactly $1/4$ to each branch, so the routing-balance regulariser is zero at the optimum of the classification objective instead of fighting it.

E. FS2K preparation for Task 4

FS2K provides 2,104 photograph-sketch pairs with an official division into 1,058 training and 1,046 test pairs and a style label in $\{0, 1, 2\}$ for each pair [35]. Pairs are matched by image number rather than sorted position, since the identifiers are not contiguous, and all 2,104 pairs were verified to be one-to-one. Reserving 15% of the official training portion as a validation split, stratified by style with seed 42, gives **899 training**, **159 validation** and **1,046 test** pairs, with styles 303/298/298, 54/52/53 and 619/381/46 respectively.

Two measured properties of FS2K shape the Task 4 experiment design.

First, **the official test set is strongly style-imbalanced**: only 46 of 1,046 test pairs (4.4%) are style 3, against 619 of style 1. Pooled metrics are therefore dominated by style 1, and per-style results are mandatory; any metric requiring a distribution estimate, such as FID, is meaningless on 46 images.

Second, and more subtly, **style is confounded with photograph source in the official training split**. FS2K draws its photographs from three sources, and in the training portion styles 1 and 2 come only from `photo1` (CASIA-WebFace [40], 250×250 tightly cropped faces) while style 3 comes only from `photo2` and `photo3` (studio cut-outs and stock photographs). A generator could therefore learn part of “style 3” from the appearance of the photograph rather than from the condition label. The official test set contains combinations never seen in training—179 `photo3` photographs with style-1 sketches—which makes it a direct test of whether the style condition generalises to a new photograph domain, exactly the situation the deployed application faces. Test metrics are therefore reported per style *and* separately for seen and unseen (source, style) combinations. A related artefact: the `photo3` style-3 sketches were drawn on grey paper (90th-percentile grey level 232 against 250–252 elsewhere), so the 179 `photo3` style-1 test pairs carry a background offset of roughly ten grey levels relative to what training taught. The sketches are left unmodified so that metrics are computed against the official ground truth, and the effect is reported rather than normalised away.

All FS2K sketches are exactly greyscale ($R = G = B$ in every pixel of all 2,104 sketches), so they are stored with one channel. Photographs and sketches are made EXIF-upright, composited onto white where transparent, centre-square cropped and resized to 128×128 with antialiased bicubic resampling; values are scaled to $[-1, 1]$ because the generator ends in a $\tanh$. The centre crop is used rather than an anisotropic resize or padding because the non-square sources are exactly the style-3 sources, so either alternative would add a second shortcut correlated with the style label. Spatial augmentation (paired horizontal flip, and the `pix2pix` jitter of upscaling by up to 286/256 and cropping back) is applied to the photograph and sketch as one stacked tensor, so identical geometry is guaranteed by construction.

IV. TASK 1: UNIVERSAL DENOISING AUTOENCODER

A. Methodology

1) *Architecture*: The universal model (`src/models/autoencoder.py`, shared with Tasks 2 and 3) is a convolutional autoencoder. A stem of two 3×3 convolution-BatchNorm-ReLU layers is followed by d stride-2 blocks that halve the resolution and double the channel count, capped at eight times the base width; a 1×1 convolution projects the deepest feature map to c_{lat} channels, and a mirrored decoder upsamples back to full resolution and ends in a sigmoid.

The depth is fixed at $d = 4$, giving an 8×8 latent. The reason is the occlusion class: the largest test occlusion is a single rectangle of roughly 76×76 pixels, and to synthesise the centre of that hole the network must see past its border on both sides. Measuring the receptive field by back-propagating from a single output pixel gives 175 px at $d = 4$ —the whole image—against 87 px at $d = 3$, which barely spans the hole.

The latent is the **only** path from input to output, asserted by a unit test that checks the output is invariant to changes in the input once the latent is held fixed. Its dimension is $c_{\text{lat}} \times 8 \times 8$ against the $3 \times 128 \times 128 = 49,152$ values of the input, so the compression ratio ranges from $96\times$ at $c_{\text{lat}} = 8$ to $12\times$ at $c_{\text{lat}} = 64$, where the search range deliberately stops: wider latents would keep improving PSNR, since more capacity always helps a pixel metric, but the brief requires a *meaningful* bottleneck.

Upsampling uses bilinear resize followed by convolution rather than transposed convolution. Transposed convolution suffers “uneven overlap” whenever the kernel size is not divisible by the stride, and the unevenness compounds across the two spatial axes into the familiar checkerboard pattern; resize-convolution avoids it because it is implicitly weight-tied in a way that discourages high-frequency artefacts [41]. (An alternative fix initialises sub-pixel convolution to be equivalent to nearest-neighbour resize-convolution [42], [43]; since the decoder here convolves after every upsampling step at every scale, the simpler resize-convolution was used.)

2) *Skip connections*: The main model has **no skip connections**. Unrestricted skips, as in U-Net [11], would let the decoder copy the input around the bottleneck, which the brief rules out; and for this task they are actively harmful, because here the input *is* the corruption—a full-resolution skip hands salt-and-pepper impulses and black occlusion rectangles straight to the decoder. Since skips are nonetheless standard in restoration, and RED-Net attributes real gains to them [12], their effect is measured rather than assumed. Three ablation variants add exactly one skip, at 16×16 , 32×32 and 128×128 respectively, with identical hyperparameters and epoch budget. Even the 16×16 skip carries $256 \times 16 \times 16 = 65,536$ values, more than the input itself, which makes the point that a “limited” skip is limited in *resolution*, not in capacity: deep features have passed through many nonlinearities and carry context, whereas a 128×128 skip is two convolutions from raw pixels.

Two diagnostics quantify the trade-off directly. *Impulse survival*, on salt-and-pepper entries, is the fraction of impulse-hit pixels whose output value is still closer to the impulse than to the clean value; it is 1 for the identity and 0 for perfect restoration. *Detail ratio*, on clean entries, is the mean absolute Laplacian of the output luminance divided by that of the target; below 1 means the output is smoother than the clean image, i.e. detail was lost in the bottleneck. Together they separate the two competing effects: skips should raise the detail ratio towards 1 and, if they leak, also raise impulse survival.

3) *Loss*: The loss is the one prescribed by the brief,

$$\mathcal{L}_{\text{UDAE}} = \alpha \mathcal{L}_{L_1}(x, \hat{x}) + (1 - \alpha)(1 - \text{SSIM}(x, \hat{x})), \quad (2)$$

with SSIM computed as specified by Wang et al. [44]: an 11×11 circularly symmetric Gaussian window of standard deviation 1.5 samples, $K_1 = 0.01$, $K_2 = 0.03$, over the valid region and averaged across channels. The L_1 term keeps colours and brightness correct; the SSIM term rewards local structure and contrast. The motivation for mixing them is Zhao

et al.’s systematic comparison, which finds that L_2 produces “splotchy” artefacts in flat regions because it tolerates small errors regardless of local structure, that L_1 converges to a better optimum, and that SSIM-type losses are insensitive to uniform biases and therefore permit colour and brightness shifts if used alone [45]. Their recommended mixture uses $\alpha = 0.84$ on the *MS-SSIM* term, chosen so the two contributions are roughly balanced; in the brief’s parameterisation, where α weights L_1 , that corresponds to $\alpha \approx 0.16$. The two conventions must not be confused, and α is tuned rather than assumed, over a range that contains both the brief’s 0.8 and Zhao et al.’s balance point. Single-scale SSIM is used rather than MS-SSIM [46] because at 128×128 a five-level pyramid would reduce the coarsest scale to 8×8 , below the 11×11 window. A perceptual, feature-space loss [47] was also considered and rejected: it optimises a quantity different from the PSNR and SSIM that are reported, and adds a VGG forward pass to every training step on a time-limited GPU.

4) *Training procedure*: Optimisation uses AdamW [48], which applies weight decay to the weights directly rather than through the adaptive gradient, with a fixed decay of 10^{-4} ; overfitting is not the principal risk when every epoch sees freshly drawn corruptions. The schedule is one epoch of linear warm-up, which avoids large early Adam steps while the second-moment estimates and the BatchNorm statistics are still noisy [49], followed by per-step cosine decay to 1% of the peak rate [50]. Mixed precision (fp16 autocast with gradient scaling [51]) is used on the GPU, since the T4 has fp16 tensor cores but no bf16 support, with SSIM always evaluated in float32 because its variance terms lose all precision in fp16. Gradients are clipped at norm 1.0 as a safety net, and a non-finite epoch loss aborts the run (or prunes the Optuna trial) rather than silently producing NaNs. Spatial dropout [52], [53] after each convolution block is searched but expected to be of little use, since runtime corruption is already a strong regulariser.

Model selection and the Optuna objective use a validation score independent of the tuned loss weights,

$$\text{score} = 0.5 \text{ SSIM} + 0.5 \frac{\text{PSNR}}{40}, \quad (3)$$

so that trials with different α remain comparable—otherwise the search would partly be optimising its own yardstick. Both components are reported separately as well, since PSNR and SSIM measure different things and can disagree [54].

5) *Optuna search design*: The study `task1_udae` maximises (3) on the validation manifest with a TPE sampler [55], [56] seeded at 42 after ten random start-up trials, and a median pruner [57] that is inactive for the first five trials and the first two epochs of each trial. Trial 0 is the brief’s own starting configuration, so the report can state how much the search gained over it. The search space is given in Table III.

Trials run 15 epochs, a quarter of the 60-epoch final run, and the cosine schedule anneals completely within each trial so that configurations are compared as fully annealed models rather than mid-schedule. The known limitation of this low-

TABLE III
OPTUNA SEARCH SPACE FOR TASK 1 (STUDY `TASK1_UDAE`): 40 TRIALS OF 15 EPOCHS EACH, TPE SAMPLER, MEDIAN PRUNER.

Parameter	Range	Scale / rationale
Learning rate	10^{-4} – 3×10^{-3}	log-uniform, brackets 10^{-3}
Batch size	{16, 32, 64}	184/92/46 steps per epoch
Latent channels	{8, 16, 32, 64}	$96 \times 12 \times$ compression
Base channels	{16, 32, 48, 64}	1.0–16.5 M parameters
Dropout	0.0–0.3	spatial dropout after each block
α (loss weight)	0.1–0.95	spans SSIM- and L_1 -dominated

TABLE IV
TASK 1 TEST RESULTS PER INPUT CONDITION ON THE 36,690-ENTRY TEST MANIFEST. CLEAN-INPUT ROWS FOR THE IDENTITY AND ORACLE BASELINES ARE EXACT, AND ARE REPORTED AS SUCH RATHER THAN AS A CAPPED PSNR.

Condition	Identity	Oracle classical	UDAE
Clean	–	–	–
Salt-and-pepper	–	–	–
Gaussian blur	–	–	–
Occlusion	–	–	–
Corrupted (pooled)	–	–	–

[TODO: fill with PSNR (dB) / SSIM from
outputs/task1/tables/comparison_by_type.tex]

fidelity protocol is that settings which pay off only late—a lower learning rate, or dropout—can be under-rated; [TODO: state whether the selected configuration differed systematically from trial 0, and whether the final 60-epoch run confirmed the 15-epoch ranking].

6) *Baselines*: Two reference points accompany every Task 1 result. The *identity* baseline is the corrupted input itself and measures how damaging each corruption is. The *oracle classical* baseline applies, to each entry, the textbook method for its corruption *using information the network never receives*: a 3×3 median filter for salt-and-pepper [58], [59]; unsharp masking with the true kernel (k, σ) , $\hat{x} = y + (y - h * y)$, which is one step of Van Cittert-style iterative deconvolution [60], [61]; and Telea fast-marching inpainting of the true rectangle mask [62], [63], [64]. Clean inputs are returned unchanged, so the oracle is exact there. Because the oracle is given the corruption type and its exact parameters, beating it is a strong result and losing to it on one corruption is an informative finding rather than a failure to conceal.

B. Results

[TODO: paragraph interpreting Table IV: which corruption the single model handles best and worst, whether it beats the oracle baselines that know the corruption, and by how much]

1) *Skip-connection ablation*: [TODO: paragraph interpreting Table VI: whether a limited skip improves clean and mildly corrupted inputs, whether impulse survival and the detail ratio confirm the predicted leak-versus-detail trade-off, and whether

TABLE V
TASK 1 TEST RESULTS PER CORRUPTION AND SEVERITY LEVEL.

Condition	Low	Medium	High
Salt-and-pepper	—	—	—
Gaussian blur	—	—	—
Occlusion	—	—	—

[TODO: fill with PSNR / SSIM from
outputs/task1/tables/by_type_level.tex; comment
specifically on the high level, which sits at the top edge of the training
ranges (Section III)]

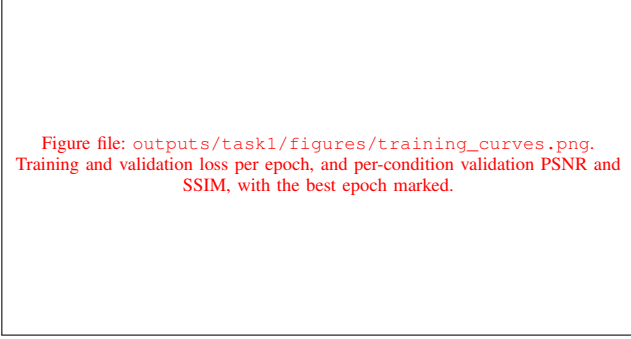


Fig. 2. Task 1 training and validation curves.

the evidence supports keeping the pure bottleneck as the main model]

C. Failure analysis

[TODO: discussion of at least four meaningful failure cases: detail lost through the bottleneck on clean inputs, residual impulses or texture confusion on heavy salt-and-pepper, unrecoverable high-frequency content on the strongest blur, and implausible inpainting of large occlusions. State for each whether the failure is a limitation of the bottleneck, of the training distribution (Section III), or of the single-model design.]

V. TASK 2: CORRUPTION CLASSIFICATION AND HARD ROUTING

A. Methodology

1) *Classifier*: The corruption classifier (src/models/classifier.py) maps an RGB image to four logits. It is a conventional convolutional classifier [65], [66] with batch normalisation [67]: each stage applies one or two 3×3 convolution-BatchNorm-ReLU layers followed by 2×2 max pooling, and two design choices follow directly from what the cues look like. The first stage runs at full 128×128 resolution, because the evidence is high-frequency: a few isolated impulse pixels, or the *absence* of high frequencies under mild blur, would both be destroyed by early downsampling. The head concatenates global average and global max pooling, because the cues are sparse—a handful of bright pixels, or one black rectangle—and average pooling alone dilutes them.

Training uses plain multiclass cross-entropy, as the brief requires, with no label smoothing: the probabilities are displayed

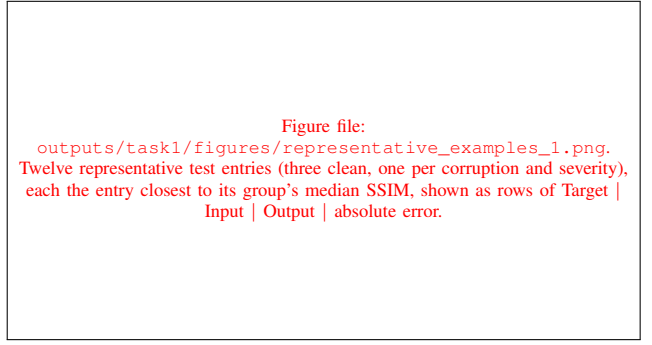


Fig. 3. Representative Task 1 restorations: clean target, corrupted input, reconstruction and absolute error map. Entries are selected by median SSIM within their group, not by hand.

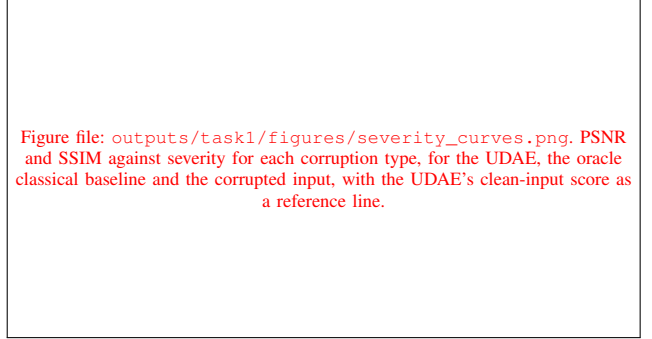


Fig. 4. Restoration quality against severity.

in the application and are used to initialise the Task 3 gate, so they should be the model's own estimates. Batches are exactly class-balanced by the sampler of Section III, and the labels come free from the corruption pipeline. AdamW is used with weight decay applied only to convolution and linear weights, not to BatchNorm parameters or biases, where decaying them does not regularise what the normalised layer computes [68].

Model selection, early stopping and the Optuna objective all use validation **macro-F1** rather than loss or accuracy. Routing uses the `argmax`, so the decision metric is what matters; macro-averaging weights the four classes equally and penalises a class that is over-predicted, which accuracy on a balanced set would hide [69]. Validation cross-entropy is logged but not used for selection, because it can rise through growing over-confidence while accuracy still improves [70].

2) *Specialists*: Three specialist autoencoders are trained, one per corruption, each seeing *only* its own corruption with a fresh severity on every load and the clean image as target. The three runs are fully independent—different seeds, separate optimisers, separate checkpoints—so their parameters are independently trained as the brief requires. They share the architecture of Section IV with a pure bottleneck and no skips, for the same reason: a specialist is still a denoising autoencoder, and a skip would pass its own corruption's high-frequency signature past the latent.

To keep the experiment computationally feasible the brief permits one shared architecture search. Each trial trains all three specialists with the trial's hyperparameters, *interleaved*

TABLE VI

SKIP-CONNECTION ABLATION: ONE LIMITED SKIP ADDED AT THE STATED RESOLUTION, SAME HYPERPARAMETERS AND EPOCH BUDGET AS THE MAIN MODEL.

Variant	Skip	Params	PSNR	Impulse surv.	Detail ratio
udae	—	—	—	—	—
udae_skip16	16 ²	—	—	—	—
udae_skip32	32 ²	—	—	—	—
udae_skip128	128 ²	—	—	—	—

[TODO: fill from outputs/task1/tables/skip_ablation.tex]

Figure file: outputs/task1/figures/failure_cases.png. Six failure modes: the lowest-SSIM output for each input condition, plus the two corrupted entries where the model gains least PSNR over its own input.

Fig. 5. Task 1 failure cases.

epoch by epoch, and reports the running mean of the three validation scores to the pruner. Interleaving matters: training them sequentially would show the pruner only the salt-and-pepper score for the first third of each trial, so a configuration that suits impulse noise but fails on occlusion would survive longer than it deserves. After the study the three specialists are retrained independently on the full schedule.

3) *Hard routing*: At inference the classifier produces $p = \text{softmax}(C(\tilde{x}))$ and the route is $r = \arg \max_k p_k$. The restorer starts from a copy of the input and overwrites only the entries with $r > 0$ using the corresponding specialist, which has three consequences worth stating: a clean-routed input is returned *bit-identical* to the input through the identity bypass, never unnecessarily processed by a restoration expert; each expert runs at most once per batch on its own sub-batch rather than once per sample; and all models run in evaluation mode, so BatchNorm uses running statistics and a sample’s output does not depend on the rest of its batch.

The system is evaluated in both modes the brief prescribes. *Oracle routing* takes r from the manifest label and isolates the restoration ability of the specialists; *predicted routing* uses the classifier and measures the complete operational system. Correctly routed entries give identical metrics in both modes, which is asserted by a test.

4) *Optuna search design*: Two studies are run (Table VII): `task2_classifier` (30 trials of 12 epochs, objective validation macro-F1) and `task2_specialists` (20 trials of 6 epochs per specialist, objective the mean of the three best validation restoration scores). Both use the TPE sampler seeded at 42 with a median pruner.

TABLE VII
OPTUNA SEARCH SPACES FOR TASK 2.

Parameter	Range	Rationale
<i>Classifier</i> (<code>task2_classifier</code>)		
Learning rate	10^{-4} – 3×10^{-3}	log-uniform
Batch size	{32, 64, 128}	multiples of 4 for exact balance
Channels	four presets	width against depth at matched cost
Convs per stage	{1, 2}	depth; also the Task 3 gate cost
Dropout	0.0–0.5	head regularisation
Weight decay	10^{-5} – 10^{-1}	log-uniform, decoupled
<i>Specialists</i> (<code>task2_specialists</code>)		
Learning rate	10^{-4} – 3×10^{-3}	log-uniform
Batch size	{16, 32, 64}	184/92/46 steps per epoch
Base channels	{16, 32, 48}	0.7–5.8 GMAC per image
Bottleneck	six latent shapes	8 ² or 16 ² by 16–128 channels
α (loss weight)	0.4–0.95	around the brief’s 0.8

TABLE VIII

TASK 2 CLASSIFIER ON THE TEST MANIFEST (36,690 ENTRIES, BALANCED BY CONSTRUCTION, SO ACCURACY EQUALS MICRO-AVERAGED F1).

Class	Precision	Recall	F1	Support
Clean	—	—	—	3,669
Salt-and-pepper	—	—	—	11,007
Blur	—	—	—	11,007
Occlusion	—	—	—	11,007
Macro average	—	—	—	36,690

[TODO: fill from outputs/task2/tables/classifier_test_per_class.tex; also state overall accuracy]

B. Results

[TODO: paragraph interpreting Figure 6: which confusions dominate, and whether they are concentrated at the low severity levels]

[TODO: paragraph on the two expected confusion mechanisms, with numbers: (i) naturally soft or low-resolution clean photographs being called blurred, quantified against the variance of the Laplacian as a focus measure [71] with Wilson intervals [72]; (ii) dark image content being called occlusion, and black occlusion rectangles on already-dark fur going undetected]

Clean inputs and the PSNR cap. Under oracle routing every clean input passes through the identity bypass, so the output equals the target exactly: zero MSE, SSIM of one, and a PSNR that is infinite and stored at the implementation’s 100 dB cap. These entries are reported as “exact (n/n)” rather than averaged as 100 dB, and the pooled restoration figures use the *corrupted* rows only. Under predicted routing the share of clean inputs that remain exact equals the classifier’s clean recall, and misrouted clean photographs receive a finite PSNR that is reported separately.

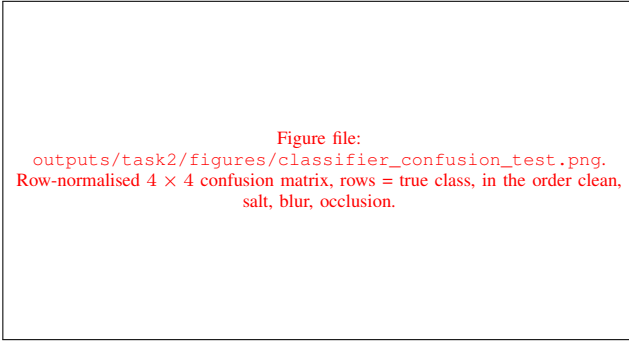


Fig. 6. Normalised confusion matrix of the corruption classifier on the test manifest. Rows are true classes and sum to one, so each row is a recall profile; this is the same normalisation used for the Task 3 routing matrix, which makes the two directly comparable.

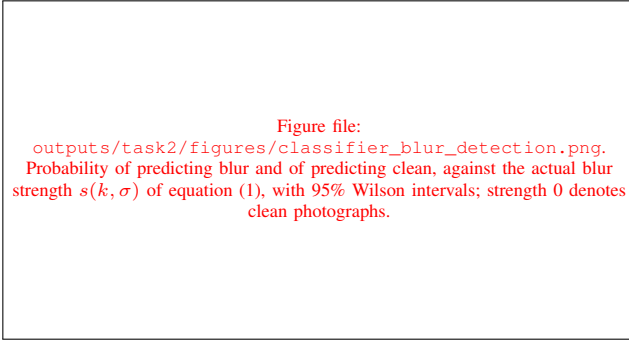


Fig. 7. Clean-versus-blur discrimination as a function of the true blur strength. The analysis uses the applied-kernel strength rather than σ , for the reason given in Section III.

C. Failure analysis

[TODO: discussion of the mis-routing table (outputs/task2/tables/routing_misrouting.tex): for each (true, routed-to) pair, how many entries, at which severities, and the mean PSNR change relative to oracle routing. Separate harmful misroutings (≥ 1 dB worse) from neutral and *beneficial* ones: a mild blur routed to the identity can score better than the blur specialist’s reconstruction if the bottleneck loses more detail than the blur itself removed. Report beneficial misroutings explicitly rather than omitting them.]

VI. TASK 3: SOFT MIXTURE-OF-EXPERTS RESTORATION

A. Methodology

1) *Model*: The soft mixture replaces the `argmax` of Task 2 with a temperature-scaled softmax over the same four branches:

$$w = \text{softmax}\left(\frac{G(\tilde{x})}{\tau}\right), \quad (4)$$

$$\hat{x} = w_0 \tilde{x} + w_1 A_{\text{salt}}(\tilde{x}) + w_2 A_{\text{blur}}(\tilde{x}) + w_3 A_{\text{occ}}(\tilde{x}). \quad (5)$$

The gate G is the Task 2 classifier and the three experts are the Task 2 specialists, loaded from their checkpoints and rebuilt from their own stored configurations, so whatever architecture

TABLE IX
 HARD-ROUTED RESTORATION: ORACLE ROUTING (TRUE LABEL) AGAINST
 PREDICTED ROUTING (CLASSIFIER), PER CORRUPTION TYPE.

Condition	Route acc.	PSNR _{or}	PSNR _{pr}	Δ	SSIM _{pr}
Clean	—	exact	—	—	—
Salt-and-pepper	—	—	—	—	—
Blur	—	—	—	—	—
Occlusion	—	—	—	—	—

[TODO: fill from
 outputs/task2/tables/routing_oracle_vs_predicted.tex]

Figure file: outputs/task2/figures/routing_failures.png. Worst
 harmful misroutings per confusion pair, showing target, input, oracle output,
 predicted output and the four classifier probabilities.

Fig. 8. Cases where a classifier error causes a restoration failure.

the Task 2 search selected is used unchanged. Branch 0 is a *true identity*, not a learned clean expert: a clean input needs only the gate to place its weight on branch 0 to be reproduced perfectly, so there is no reconstruction floor for clean images.

The temperature τ is a registered buffer rather than a parameter or a Python float, which means it is saved with the state dictionary, is never touched by the optimiser, and is traced into the exported ONNX graph as a constant, so the deployed model provably uses the temperature it was trained with. Temperature scaling of a softmax follows Hinton et al. [73]: a higher T produces a softer distribution. Here a small τ approaches the hard routing of Task 2 and a large τ approaches uniform blending, so τ is the single knob that interpolates between Tasks 2 and 3—which is why it is searched. (A stochastic hard route with a differentiable relaxation, as in the Gumbel-softmax [74], was the alternative not taken: the brief asks for a continuous weight on every branch, and a dense blend needs no relaxation.) The gate logits and the softmax are computed in float32 even under mixed precision, since the routing decision multiplies *every* branch output.

A caveat that should be stated plainly. Equation (5) trained on the error of the blend is exactly the *cooperative* error function of Jacobs et al. [14], the formulation they argue “does not encourage localization” because each expert need only cancel the residual left by the others. Two features of this system counteract that. The experts are pre-trained to specialise rather than learned from scratch, and the cross-entropy term below supervises the routing directly with the known runtime corruption label. Without both, the blend objective would be expected to drift towards using many experts for every input.

2) Loss:

$$\mathcal{L}_{\text{MoE}} = \lambda_1 \mathcal{L}_{L_1} + \lambda_s (1 - \text{SSIM}) + \lambda_c \mathcal{L}_{\text{CE}} + \lambda_b \mathcal{L}_{\text{balance}}. \quad (6)$$

The cross-entropy is applied to the *raw* gate logits $G(\tilde{x})$ rather than to $G(\tilde{x})/\tau$. With the scaled logits, the same parameter τ would control both the sharpness of the routing and the magnitude of the classification gradient—a small τ multiplies the cross-entropy gradient by $1/\tau$ —so the searches over τ and λ_c would be confounded. Keeping the cross-entropy on the raw logits makes τ purely a routing knob, λ_c purely a weight, and preserves the Task 2 objective so that the gate remains a calibrated corruption classifier whose weights are interpretable in the gating analysis and in the application display.

3) *The balance regulariser*: The regulariser is the brief’s,

$$\mathcal{L}_{\text{balance}} = \sum_{k=0}^3 \left(\bar{w}_k - \frac{1}{4} \right)^2, \quad (7)$$

with $\bar{w}_k$ the mean weight of branch k over the batch. Three arguments support it, and the alternatives were examined in detail before it was kept.

First, it is the appropriate simplification for a dense gate. Shazeer et al. use two terms [17]: an importance loss $\mathcal{L}_{\text{imp}} = w_{\text{imp}} \cdot \text{CV}(\text{Importance}(X))^2$, the squared coefficient of variation of the batchwise sum of gate values, and a load loss $\mathcal{L}_{\text{load}} = w_{\text{load}} \cdot \text{CV}(\text{Load}(X))^2$ over a smooth estimator of the *number of examples* routed to each expert. The Switch Transformer merges them into

$$\mathcal{L}_{\text{switch}} = \alpha N \sum_{i=1}^N f_i P_i, \quad (8)$$

where f_i is the fraction of tokens dispatched to expert i and P_i the mean router probability, with $\alpha = 10^{-2}$ chosen after a sweep over five orders of magnitude [18], [19]. Both the load term and f_i exist to count *discrete dispatches* in a top- k router. The gate here is dense: all four branches always run, nothing is dropped, and there is no load to count, only importance. With soft routing f and P coincide, and since $\sum_k (P_k - 1/N)^2 = \sum_k P_k^2 - 1/N$, equation (7) is the Switch loss up to an affine transformation. The brief’s term is therefore not a weaker substitute for the published losses; for a dense gate it is the same objective.

Second, it does not fight the supervised objective. Because training batches contain exactly $B/4$ images of each condition, perfect routing—one-hot on the true class—gives every branch a mean weight of exactly $1/4$, so $\mathcal{L}_{\text{balance}} = 0$ at the optimum of the cross-entropy. The two terms are compatible by construction, a property that follows from the balanced sampler and is unavailable in the MoE literature, which has no labels. (With an unbalanced batch the term would actively punish correct routing, which a unit test asserts.) Total collapse onto a single branch costs $(1 - 1/4)^2 + 3 \cdot (1/4)^2 = 0.75$.

Third, the exact form matters less than its presence. Shazeer et al.’s own ablation, reproduced in Table X, shows that any combination containing at least one balance term gives

TABLE X

EFFECT OF THE BALANCE LOSSES IN SHAZEER ET AL. [17], TABLE 6 (MoE-256 MODEL, 10 EPOCHS). ANY BALANCE TERM SUFFICES; NONE IS MUCH WORSE. CV DENOTES THE COEFFICIENT OF VARIATION.

w_{imp}	w_{load}	Perplexity	CV(Imp.)	CV(Load)	$\frac{\text{max}}{\text{mean}}$
0.0	0.0	39.8	3.04	3.01	17.80
0.2	0.0	35.6	0.06	0.17	1.47
0.0	0.2	35.7	0.22	0.04	1.15
0.1	0.1	35.6	0.06	0.05	1.14
0.01	0.01	35.7	0.48	0.11	1.37
1.0	1.0	35.7	0.03	0.02	1.07

nearly identical model quality over two orders of magnitude of weight, whereas omitting them entirely is far worse.

Entropy regularisers were considered and rejected as a *loss*. Maximising routing entropy would push against the cross-entropy by rewarding weight spread onto wrong branches; minimising it would push towards hard routing, which is Task 2. (The information-maximisation combination of a high marginal entropy and a low conditional entropy [75], [76] remains the principled upgrade if a stronger regulariser is ever needed, and confidence penalties [77] the remedy if the gate saturates.) Entropy is used here as a *diagnostic* instead, reported per class. The one blind spot of (7) is that it sees only batch-mean weights and is therefore insensitive to a gate that routes each class confidently to the *wrong* branch; the cross-entropy term and the routing matrix below exist for that.

4) *Routing collapse*: Collapse is defined once and used identically by the Optuna pruner, the per-epoch log and the final evaluation. From the routing matrix on the balanced validation manifest, $R[c, k]$ being the mean weight of branch k over entries of true class c (rows sum to one), define $\text{usage}_k = \text{mean}_c R[c, k]$, which is $1/4$ under balanced perfect routing, and $\text{off}_k = \text{mean}_{c \neq k} R[c, k]$. A configuration has collapsed if some branch is *dead* ($\text{usage}_k < 0.05$, a fifth of its fair share; since $\text{usage}_k \geq R[k, k]/4$ such a branch receives less than 0.2 of the weight even on the inputs it exists for) or *dominating* ($\text{off}_k > 0.5$, taking more than half the weight on average on the other three classes). The criterion deliberately does not flag a merely uncertain model: uniform routing is informative, not useless, and the tests assert that uniform weights are *not* flagged while total collapse is flagged as both dead and dominating.

5) *Staged training*: Training proceeds in the two stages the brief prescribes. In the **warm-up** the experts are frozen in two senses at once: their parameters have `requires_grad = False` and the warm-up optimiser is built over the gate only, and they are kept in evaluation mode with the `train()` method overridden so that a later call cannot silently undo it. The second part is not redundant—without it, BatchNorm would keep updating its running statistics on every forward pass and dropout would stay active, so the “frozen” experts would drift and the warm-up would no longer be a clean measurement of the gate. In the **joint stage** all parameters train, with the gate at learning rate η and the experts at 0.1η , under cosine decay.

TABLE XI
OPTUNA SEARCH SPACE FOR TASK 3 (STUDY `TASK3_MOE`).

Parameter	Range	Rationale
Joint learning rate	10^{-5} – 3×10^{-4}	log; capped at the warm-up rate so the joint stage stays gentler
Temperature τ	0.3–3.0	log; near-hard to near-uniform routing
λ_c	0.01–1.0	log; a decade either side of 0.1
λ_b	10^{-3} – 10^{-1}	log; contains Switch’s $\alpha = 10^{-2}$
α ($\lambda_1 = \alpha$, $\lambda_s = 1 - \alpha$)	0.5–0.95	keeps the reconstruction scale constant

The schedule’s shape has direct empirical support. Kumar et al. [78] show that full fine-tuning from a randomly initialised head distorts good pretrained features, because the lower layers move while the head is still wrong, and that linear probing followed by full fine-tuning combines the benefits of both. The warm-up is exactly the probe stage, with the gate as the head: it prevents a miscalibrated gate from pushing destructive gradients into three specialists that are already good. The lower expert learning rate in the joint stage is discriminative fine-tuning [79], and is additionally motivated by the fact that the gradient reaching an expert is scaled by its routing weight, so a rarely selected expert receives little signal while a dominant one receives a lot. The experts’ BatchNorm running statistics are kept fixed during joint training, since an expert’s input distribution changes once it sees all four conditions and re-estimating would move its normalisation away from the regime it was trained in [80].

The validation objective caps per-image PSNR at 40 dB *inside the objective only*, because a clean input routed to the identity reaches 60–100 dB and would otherwise dominate model selection over actual restoration quality. The uncapped score is logged, and reported test metrics are never capped this way.

6) *Optuna search design*: The study `task3_moe` searches the five parameters the brief names (Table XI), with trial 0 enqueued at the brief’s starting values $\lambda_1 = 0.8$, $\lambda_s = 0.2$, $\lambda_c = 0.1$, $\lambda_b = 0.01$. Trials run one warm-up and four joint epochs. Pruning has two paths, both recorded in the trial attributes: the median pruner for weak configurations, and the collapse criterion above—a task-specific stopping rule that no generic pruner can express, and which the brief explicitly permits. A diverged loss is converted into a prune with its reason rather than a crashed trial, so one bad learning rate cannot end a session.

B. Results

1) *Gating analysis*: [TODO: paragraph interpreting Figures 9–11 and Table XIV: how routing weight varies with severity (are mild corruptions routed more softly?), whether any expert became inactive or dominates unrelated inputs, what the selected τ implies about where the system sits

TABLE XII
SOFT MIXTURE-OF-EXPERTS RESTORATION ON THE TEST MANIFEST, PER CORRUPTION AND SEVERITY.

Condition	Low	Medium	High
Salt-and-pepper	–	–	–
Blur	–	–	–
Occlusion	–	–	–
Clean	[TODO: PSNR / SSIM and number of near-exact entries]		

[TODO: fill with PSNR / SSIM from
outputs/task3/tables/by_type_level.tex]

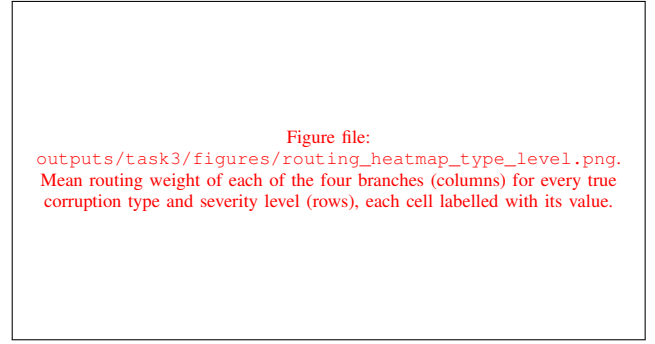


Fig. 9. Average expert weights for every true corruption type and severity level. Rows sum to one.

between hard and uniform routing, and whether the balance weight selected by Optuna was large enough to matter]

C. Failure analysis

[TODO: discussion of the worst mis-routed entries (outputs/task3/figures/), of whether soft routing recovers cases where the Task 2 classifier was confidently wrong, and of cases where blending made the result worse than hard routing would have. Include the optional two-corruption experiment (manifests/pets_mixed_manifest.json), clearly labelled as beyond the brief’s corruption definitions, reporting the mean weights per combination and the share of entries whose two largest weights are exactly the two corruptions present.]

VII. TASK 4: STYLE-CONDITIONED FACE-TO-SKETCH GENERATION

A. Methodology

1) *Generator*: The generator follows the `pix2pix_unet_128` layout [25]: seven 4×4 stride-2 convolutions down to 1×1 with LeakyReLU(0.2), then six transposed convolutions with skip connections and a tanh output, with dropout after the three innermost decoder layers, following the convolution-normalisation-activation modules that `pix2pix` inherits from DCGAN [81] together with its $\mathcal{N}(0, 0.02)$ weight initialisation. It is worth noting the contrast with Section IV: the U-Net skips that are *forbidden* there, because they would defeat the bottleneck, are *required* here, because the goal is a translation that preserves the input’s spatial layout rather than a compressed representation. Architectural choices follow the objective.

TABLE XIII

CROSS-TASK COMPARISON ON IDENTICAL TEST ENTRIES: THE UNIVERSAL MODEL, HARD ROUTING IN BOTH MODES, AND THE SOFT MIXTURE. PSNR COLUMNS EXCLUDE ENTRIES AT THE CAP; *exact* COUNTS THEM.

Condition	Task 1 universal		Task 2 oracle		Task 2 predicted		Task 3 soft MoE	
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
Clean	—	—	—	—	—	—	—	—
Salt-and-pepper	—	—	—	—	—	—	—	—
Blur	—	—	—	—	—	—	—	—
Occlusion	—	—	—	—	—	—	—	—
Corrupted (pooled)	—	—	—	—	—	—	—	—

[TODO: fill from `outputs/task3/tables/` produced by `src/task3/compare.py`; add the paired per-entry win rate of the soft mixture over hard predicted routing, split by whether the Task 2 classifier was correct — the case where soft routing is expected to pay off]

Figure file: `outputs/task3/figures/weight_distributions.png`.
Distribution of each branch weight within each true class.

Fig. 10. Routing-weight distributions per true class.

TABLE XIV

EXPERT ACTIVITY: IS ANY BRANCH INACTIVE, AND DOES ANY BRANCH DOMINATE UNRELATED INPUTS?

Branch	Usage	Own class	Off-class	Worst off-class
Identity	—	—	—	—
Salt-and-pepper	—	—	—	—
Blur	—	—	—	—
Occlusion	—	—	—	—

[TODO: fill from `outputs/task3/tables/expert_activity.tex`; state the collapse verdict and the mean normalised routing entropy]

Instance normalisation is used rather than batch normalisation. `pix2pix`'s batch normalisation with batch size one and test-batch statistics is instance normalisation in all but name [25], [82]; making it explicit means training and evaluation behave identically—so ONNX inference matches training—and makes the network independent of the batch size, which Optuna tunes.

Style conditioning uses a single learned embedding of the three styles. At each of the six decoder layers, a conditional instance normalisation computes $\text{IN}(h) \cdot (1 + \gamma_l(e)) + \beta_l(e)$, where γ_l and β_l are linear projections of the shared embedding e . This is FiLM in the form of conditional instance normalisation [28], [27], [29]; projecting one shared embedding to every layer's gains and biases follows BigGAN's shared embedding [83], and modulating the decoder is what StarGAN v2 does for multi-domain translation [84], [31]. The encoder stays style-agnostic, since it only has to read the photograph.

Figure files: `outputs/task3/figures/gating_dominant.png` and `gating_spread.png`. Entries with the lowest and highest routing entropy per class, each showing all four branch outputs with their weights, the mixed output and the target.

Fig. 11. Inputs on which one expert dominates (low routing entropy) and inputs on which the weights are distributed across several experts (high entropy). Examples are selected by entropy rather than by hand, so they are reproducible.

The alternatives were not merely argued but measured in a controlled experiment: one synthetic photograph with three target sketches differing only in style, the same U-Net, two seeds, comparing the L_1 error under the true style against the error under a shifted style. A style-blind model can only reach the median of the three targets, around 0.054. Input concatenation scored 0.0558 with a gap of *exactly zero*—it learned nothing from the label, as the theory predicts: a spatially constant input map is only a conditional bias on the first layer, and the instance normalisation of the next layer subtracts per-channel offsets, so the signal is normalised away [29]. Injecting at the 1×1 bottleneck gave a gap of 0.0026, limited because the U-Net skips let the decoder bypass the bottleneck entirely. Decoder FiLM gave 0.0452 against 0.0555, a gap of 0.0104—four times the bottleneck variant—and also modulating the encoder added nothing measurable (0.0108). Decoder FiLM was therefore chosen.

2) *Discriminator*: The discriminator is the `pix2pix` 70×70 PatchGAN on the concatenation of photograph and sketch, with three downsampling layers, instance normalisation and LeakyReLU(0.2), producing a 14×14 map of logits for 128×128 inputs. The 70×70 receptive field was verified by gradient propagation with the normalisations disabled. `pix2pix` found this patch size best among 1, 16, 70 and 286 pixels at 256×256 resolution: a 16×16 patch is enough for sharpness but tiles, and the full-image discriminator scored considerably

worse [25]. At 128×128 a 70-pixel patch covers about half a face.

Style conditioning in the discriminator is by projection: the output layer is a shared 4×4 kernel plus a style-specific kernel generated from the discriminator’s own embedding, so each patch logit is $\text{logit}(p) = w \cdot \phi(p) + b + v(s) \cdot \phi(p)$. This is Miyato and Koyama’s projection discriminator, $f(x, y) = y^T V \phi(x) + \psi(\phi(x))$, applied per patch [33]; it follows from assuming that $p(y | x)$ is log-linear in the features, and on ImageNet it beat both concatenation and the AC-GAN auxiliary classifier [32]. Concatenation would face the same normalisation problem as in the generator. The photograph, being a genuinely spatial condition, is still supplied by concatenation as in pix2pix. Implementation detail: one convolution produces a logit map per style and a `gather` selects each row’s style, so only the used embedding rows receive gradients.

Two options are held in reserve should training prove unstable or style control weak: spectral normalisation of the discriminator [85] if it overpowers the generator, and mismatched-condition negatives—a real sketch presented with the wrong style and labelled fake, as in GAN-CLS [86]—if the source/style confound of Section III shows up as weak style control.

3) *Objective and training:* The generator objective is adversarial plus reconstruction,

$$\mathcal{L}_G = \mathcal{L}_{\text{adv}} + \lambda_{L_1} \mathcal{L}_{L_1}(y, G(x, s)), \quad (9)$$

with binary cross-entropy with logits for the adversarial terms. pix2pix uses $\lambda_{L_1} = 100$ and reports that L_1 alone gives blurry results while the conditional GAN alone gives sharp results with artefacts, the combination reducing them [25]; 100 is the starting point here and λ_{L_1} is searched. Following pix2pix, Adam [87] is used with $\beta = (0.5, 0.999)$ and the discriminator objective is halved to slow it relative to the generator; separate learning rates for the two networks are standard [88]. All spatial augmentation is applied identically to both members of a pair, as required to preserve pixel correspondence. Unlike pix2pix, dropout is active only in training mode: the application needs deterministic sketches and exact ONNX parity, so inference runs in evaluation mode.

4) *Optuna search design:* The study `task4_cgan` searches the generator and discriminator learning rates, batch size, base channel count, dropout rate, style-embedding dimension and λ_{L_1} . Because GAN training is expensive, trials use a reduced epoch budget and the selected configuration is retrained on the full schedule, as the brief permits. [TODO: complete search-space table with the exact ranges from `src/task4/optuna_search.py`, the number of completed and pruned trials, and the selected configuration]

B. Results

What these metrics can and cannot support. SSIM [44] is paired, interpretable and the metric the FS2K benchmark itself reports, alongside the sketch-specific SCOOT measure [89], [35]; since SCOOT is not implemented here, the numbers

TABLE XV
TASK 4 RESULTS ON THE OFFICIAL FS2K TEST SET (1,046 PAIRS), PER STYLE. NOTE THE SUPPORT: STYLE 3 HAS ONLY 46 TEST PAIRS.

Style	Pairs	SSIM	LPIPS	L_1
Style 1	619	—	—	—
Style 2	381	—	—	—
Style 3	46	—	—	—
Macro average	—	—	—	—
Pooled	1,046	—	—	—

[TODO: also report FID on the full test set only, with the sample size and the resize path stated; and the seen/unseen (source, style) breakdown of Section III]

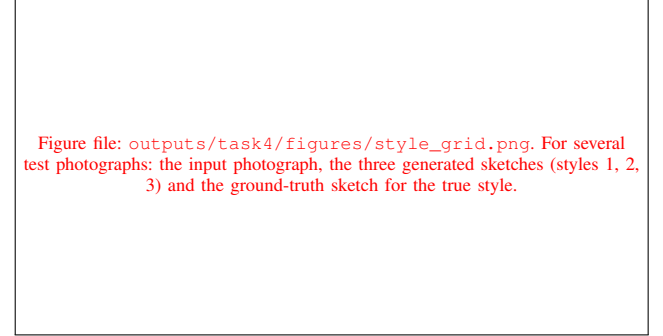


Fig. 12. The same photograph rendered in all three styles, next to the ground-truth sketch. The style condition is the only input that differs across the three generated columns.

above are *not* directly comparable with published FS2K benchmark results, and no such comparison is claimed. LPIPS [90] is included because PSNR and SSIM are shallow functions that miss much of human perceptual judgement, and because, being paired, it remains well defined on 46 images. FID [88] is reported on the full test set only and never per style: Chong and Forsyth [91] prove that FID is biased at finite sample sizes *and* that the bias depends on the model being evaluated, so one model can beat another purely through a smaller bias—an effect that dominates at $n = 46$. Parmar et al. [92] further show that low-level choices, particularly resizing and compression, induce large FID variations, so the resize path is stated explicitly and kept identical for real and generated images. KID [93] would be better behaved at small n , since it assumes no parametric form for the activations and has a simple unbiased estimator. More generally, GAN metrics disagree [94], [95], so small differences should not be over-read.

[TODO: paragraph reporting the style-sensitivity metric (validation L_1 under the true style against a shifted style) and what it says about how much the model actually uses the condition, as opposed to inferring the style from the photograph’s appearance — the confound identified in Section III]

C. Failure analysis

[TODO: failure cases: photographs far from the FS2K framing (off-centre or small faces), the unseen (source, style) combinations in the test set, the grey-paper offset affecting

Figure file: outputs/task4/figures/training_curves.png.
Discriminator real and fake losses, generator adversarial and reconstruction losses, and validation measurements, each logged separately.

Fig. 13. Task 4 training curves.

the 179 photo3 style-1 pairs, and any style for which the condition is weakly obeyed. Include generated examples.]

VIII. EXPERIMENTAL SETUP

All training runs on a single free-tier Google Colab [96] instance with an NVIDIA T4 GPU [97], using PyTorch [98] with mixed precision [51]. Because free-tier sessions disconnect without warning, every component is built to resume: training writes a complete checkpoint every epoch (model, optimiser, scheduler, gradient scaler, early-stopping state, history and the Weights & Biases run id) and `-resume` continues from it with the stored configuration treated as authoritative, so a resumed run cannot silently change its schedule. The best checkpoint is written before the last one, so a crash between the two can never leave the best behind. Data order and runtime corruptions depend only on the seed and the epoch index, so a resumed run sees the same data as an uninterrupted one; a test interrupts a run mid-epoch and asserts that the resumed weights match the uninterrupted ones exactly.

Hyperparameter optimisation uses Optuna [56], [99]. Its define-by-run interface allows the search space to be built dynamically, which is what lets conditional parameters (such as a latent shape that implies a depth) be expressed directly. All studies use the TPE sampler [55], [100] seeded at 42, which fits one density to the parameters of the best trials and another to the rest and maximises their ratio; random search would be the weaker baseline [101]. Pruning uses the median stopping rule [57], which stops a trial whose best intermediate value is worse than the median of completed trials at the same step and which Vizier reports gives a two- to three-fold speedup while still finding the best trial. Successive halving and Hyperband [102], [103], [104] were considered and not used: with 20–40 trials on one T4, their aggressive early stopping would spend most of the budget on very short runs whose ranking is unreliable for GAN and mixture training, whereas the median rule is model-free and matches the fixed-epoch validation schedule. Task 3 adds the task-specific routing-collapse criterion of Section VI, which no generic pruner can express. Each study’s SQLite database is mirrored to Google Drive after every finished trial, so a search survives a disconnect and only the remaining trials are run.

Figure file: screenshot of the original Google Stitch design (evidence of the Stitch design, required by the brief).

Fig. 14. The original Google Stitch design of the application interface.

Experiment tracking uses Weights & Biases [105]. Every Optuna trial is its own run with its parameters in the configuration and its per-epoch validation metrics logged; final runs additionally log per-condition metrics, sample image grids at fixed intervals using *fixed* validation indices so that progress is comparable across epochs, and the best checkpoint as a model artefact.

[TODO: report the actual wall-clock times, the number of completed and pruned trials per study, and the Weights & Biases project URL once the runs are finished]

IX. APPLICATION ARCHITECTURE

A. Interface design

The interface was designed in Google Stitch [106] before implementation, as the brief requires. The design defines a four-tab shell—one tab per workspace, named exactly as the brief names them—over a two-column layout: a narrow input card with an upload dropzone, sample images, corruption and severity controls and an action button, and a wide results column with image panels, a details card and a download control. Each workspace extends the shell with what its task exposes: a routing toggle and classifier probability bars for hard routing, four routing weights for the mixture, and a style selector for the sketch generator.

B. Backend

The backend is a FastAPI application [107] that runs every model through ONNX Runtime [108] on CPU. It exposes a health endpoint reporting status, uptime, runtime versions and the model card of every loaded model; an options endpoint returning the corruption types, fixed levels, permitted custom parameter ranges and style names, so the interface builds its controls from the server rather than duplicating constants; a samples endpoint; a corruption endpoint that applies a runtime corruption for preview; and one endpoint per task. Crucially, the backend imports *the same* corruption and preprocessing code as training (`src/data/`), so an image passed through the application is degraded and prepared exactly as it would have been during training.

Uploads are validated by decoding rather than by declared content type, with limits on file size and pixel count to reject decompression bombs; parameters outside the training ranges

Figure file: architecture diagram — browser (React + Tailwind) → FastAPI backend → ONNX Runtime session registry → the five ONNX models, with the shared `src/data` corruption and preprocessing code highlighted as common to training and serving.

Fig. 15. Application architecture.

Figure files: screenshots of the four workspaces — Universal Restoration, Hard-Routed Restoration, Soft Mixture-of-Experts Restoration and Face-to-Sketch Generator — each showing an input, the result and the task’s own information panel.

Fig. 16. The four application workspaces.

are refused with an explanatory message rather than silently clamped, so the evaluator cannot unknowingly test the models outside the distribution they were trained on. Requests may supply either an uploaded file or a bundled sample, and either a fixed severity level or custom parameters, never both.

C. Frontend and deployment

The frontend is a React [109] application styled with Tailwind CSS [110], implementing the four workspaces of the Stitch design against the backend API. Frontend and backend each run in their own container and start together with a single Docker Compose command [111].

D. ONNX export and parity

Five models are exported to ONNX [112], [113] at opset 17 in float32 with a dynamic batch axis: the universal auto-encoder, the classifier, the three specialists, the complete mixture pipeline as a single graph (gate, temperature constant, softmax, identity branch, three experts and the weighted sum, emitting the output, the four weights and all four branch outputs) and the sketch generator. Every export is verified against its PyTorch original in evaluation mode on at least 32 *real* test inputs spanning all conditions and severities—not random noise, so that the gate sees realistic logits—and the maximum absolute difference must stay below 10^{-4} or the file is rejected and the script fails. Each model is accompanied by a JSON model card recording its configuration, input and output specification, preprocessing, test metrics, hyperparameters, the measured parity, the Weights & Biases run URL, the git commit and the export time. [TODO: report the measured

maximum absolute difference for each of the five exported models]

X. LIMITATIONS

Severity coverage. All three test “high” levels sit at the upper edge of the training ranges, and no training blur ever reaches the strength of the (7, 2.5) test configuration (Section III). Results at the high level therefore measure extrapolation to the boundary of the training distribution as much as restoration ability.

Search fidelity. Hyperparameters were selected from short trials—15 epochs for Task 1, 12 for the Task 2 classifier, 6 per specialist, 5 for Task 3—and applied to runs three to four times longer. Short budgets favour configurations that learn quickly; the best learning rate for a longer schedule is typically somewhat lower. This is the usual trade-off of a budgeted search on a single free-tier GPU, not a claim that the selected configuration is optimal.

Scale. With 2,944 training images at 128×128 and models of a few million parameters, none of these systems is comparable with published all-in-one restoration models [2], [3], which are trained on orders of magnitude more data. No such comparison is made.

The mixture objective. As Section VI notes, training the blend on its reconstruction error is the cooperative formulation that Jacobs et al. identify as discouraging specialisation [14]. The pre-trained experts and the cross-entropy term counteract it, but the system is not an independent demonstration that soft routing induces specialisation; the specialisation was supplied by Task 2.

Balance regulariser blind spot. Equation (7) sees only batch-mean weights and cannot detect a gate that routes each class confidently to the wrong branch. The routing matrix and the cross-entropy term cover that case, but the regulariser alone does not.

FS2K confounds and sample sizes. Style is confounded with photograph source in the official FS2K training split, so part of what looks like style control may be source recognition; and the test set contains only 46 style-3 pairs, which makes distributional metrics unusable for that style and widens the error bars on every per-style number (Section III).

Metrics. PSNR correlates with subjective quality only within a fixed content set [114], SSIM and PSNR are shallow functions [90], and FID is biased in a model-dependent way at small sample sizes [91]. Conclusions are therefore drawn from several metrics together with visual evidence, never from a single number.

Deployment. ONNX inference runs on CPU in the application container, so reported inference times reflect the evaluation machine and are not comparable with GPU timings.

XI. CONCLUSION

[TODO: conclusion, to be written once results exist. It should state: which of the three restoration designs performed best overall and on which corruptions; whether explicit routing (Task 2) justified its extra complexity over one shared model

(Task 1); whether making the routing soft and differentiable (Task 3) recovered the classifier errors it was intended to, and at what cost; whether the Task 4 generator genuinely uses the style condition rather than inferring style from the photograph; and which of the research-driven design decisions (pure bottleneck, L1/SSIM mixture, resize-convolution, balance regulariser, decoder FiLM with a projection discriminator) were confirmed by the experiments and which were not.]

APPENDIX USE OF AI ASSISTANCE

AI assistants were used during this work, as the brief permits. Every generated artefact was reviewed, tested and corrected by the author, who remains responsible for the correctness of the submission. Table XVI lists each tool, what it was used for, and how its output was verified.

TABLE XVI
USE OF AI ASSISTANCE.

Tool	Used for	How it was verified
[TODO: tool, version]	[TODO: e.g. drafting training and evaluation scripts]	[TODO: e.g. unit tests; smoke runs; line-by-line review]
[TODO: tool]	[TODO: e.g. literature search and reference collection]	[TODO: every citation checked against the publisher, arXiv or proceedings page; unverifiable claims removed]
[TODO: tool]	[TODO: e.g. interface design]	[TODO: design reviewed and reimplemented by hand in React]
[TODO: tool]	[TODO: e.g. drafting report sections]	[TODO: every number recomputed from the committed manifests and result files]
[TODO: tool]	[TODO: e.g. debugging]	[TODO: fixes reproduced locally and covered by regression tests]

AVAILABILITY

The complete implementation—data preparation, training, Optuna search, evaluation and ONNX export scripts, backend, frontend, Dockerfiles and Docker Compose configuration—is available at <https://github.com/fsdev87/generative-ai-A1>. The demonstration video is at [TODO: YouTube link].

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